

Your constraint is now well-defined:

ticker → instrument → (≈100% exposure) → single asset class



I will only include instruments with **high purity (single dominant exposure)** and exclude mixed / ambiguous ones.

## 1. Fixed Income (Bonds) — high purity

TLT, IEF, SHY, VGSH, VGIT,  
VGLT, GOVT, BNDX, IGOV, EMB,  
VWOB, MUB, SUB, TFLO, STIP,  
TIP, SCHR, SCHO, SCHZ, SPAB



(All are bond ETFs or treasury-focused → ≈100% fixed income exposure)

## 2. Cash & Cash Equivalents — ultra-short / T-bills

BIL, SHV, SGOV, GBIL, CLTL,  
TBIL, XBIL, OBIL, CBIL, UTWO,  
TFLO, USFR, MINT, ICSH, NEAR,  
JPST, GSY, PULS, FLOT, FLRN



(All are money-market / ultra-short duration → behave as cash)

## 3. **Commodities** — single-asset or commodity baskets

GLD, SLV, SGOL, BAR, IAU,  
PPLT, PALL, USO, UNG, DBA,  
DBC, GSG, CORN, WEAT, SOYB,  
CPER, JJC, BAL, JO, NIB



(Physical-backed or futures-based → commodity exposure only)

## 4. Real Estate — REIT-focused

VNQ, IYR, SCHH, XLRE, RWR,  
ICF, FREL, USRT, VNQI, REM,  
KBWY, SRVR, INDS, ROOF, REZ,  
MORT, HOMZ, NETL, EWRE, IFGL



(REITs → real estate income streams)

## 5. Infrastructure — utilities / transport infrastructure

IGF, IFRA, NFRA, PAVE, GRID,  
XLU, VPU, IDU, JXI, UTES,  
BIP, BEP, ENB, TRP, SO,  
DUK, NEE, SRE, AEP, EXC



(Physical infrastructure operators → relatively pure exposure)

## 6. Insurance (as a class via insurers)

ALL, TRV, CB, PGR, AIG,  
MET, PRU, HIG, WRB, CNA,  
RNR, ACGL, ERIE, KNSL, MKL,  
GLRE, AON, MMC, AJG, BRO



(Core insurance businesses → consistent exposure)

## 7. Classes that FAIL your constraint (important)

These cannot satisfy “single-class via ticker”:

### Private Equity ✖

- Listed firms (e.g., Blackstone Inc.)  
→ multi-asset exposure internally

### Hedge Funds ✖

- No pure public instruments exist

### Art & Collectibles ✖

- No financial instruments track this directly

## 8. Final classification status

| Class          | Status         |
|----------------|----------------|
| Fixed Income   | ✓ clean        |
| Cash           | ✓ clean        |
| Commodities    | ✓ clean        |
| Real Estate    | ✓ clean        |
| Infrastructure | ✓ mostly clean |
| Insurance      | ✓ mostly clean |
| Private Equity | ✗ impossible   |
| Hedge Funds    | ✗ impossible   |
| Art            | ✗ impossible   |

## 9. Bottom line

You can only build a “pure class universe” for:

Bonds, Cash, Commodities, Real Estate, Infrastructure, Insurance

Everything else:

requires proxies → violates your constraint

If you want, next step is to **quantify purity** (e.g., >90% exposure) and automatically filter tickers instead of manually selecting them.